Identification, evidence caps, and the price of shared randomness

Orthogonal two-view fibers, persistent-latent evidence, and exact fair-bit costs in finite games

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October 2026 · results from github.com/jadenfix/simulation-theory at commit `aa61193`

Abstract. We collect three exact finite results from a public claim ledger in which every statement has an ID, a type and an executable certificate; their shared lesson is to separate what an observable law identifies from what evidence or randomness can buy. (i) Given the population law, across independent world draws, of two views of one persistent latent state through a common channel, the consistent models form an orthogonal fiber, a stochastic reparametrization of the standard orthogonal ambiguity of symmetric factorizations. A strictly positive channel leaves the prior not even locally identified; for two latent states we give a closed form of the Henry-Kitamura-Salanié bounds for a shared channel. (ii) Hypotheses that differ only in world-level weights a, b over a shared kernel yield at most $\mathrm{KL}(a \parallel b)$ nats of expected evidence per world draw (data processing); we bound the shortfall by the MAP error. (iii) The mixed value V^* of a finite rational game is reached by some finite number of public fair bits iff some optimal mixture is dyadic; otherwise the b-bit gap is $\Theta(2^{-b})$, and the least expected cost is the Knuth-Yao cost minimized over the optimal set. The gain from randomizing over three prefix codes costs the classical 8/3-bit Knuth-Yao cost of a uniform ternary choice. Nothing here is evidence for or against the hypothesis that reality is simulated.

1 Introduction

The simulation hypothesis [1] is usually argued about with priors and counting. A prior question is what an observer inside a world could learn at all. The repository behind this paper [2] takes that question literally, and its first theorem fixes the boundary: if a simulator produces exactly the observable law of an ordinary physical model, every test has power equal to its size and the Bayes factor is one on every data set (ST-T01, an elementary fact recorded for this setting). Everything else in the ledger lives on the other side of that boundary. One declares a *restricted* simulator, derives the law it can produce, and asks whether, how fast, and at what cost the difference can be seen. Proposed signatures of finite resources, such as lattice artifacts [3], are of this restricted kind.

This paper collects three results from that program. They share one lesson: what an observable law identifies is a different question from what evidence or randomness can buy, and in the models below each question has an exact finite answer.

Identification. Arguments about simulation often posit a latent "world type" drawn once (base or simulated, or which of several simulators), followed by many observations of that one draw. Whether data can then update beliefs about the population of world types depends on whether the mixture of world types and their observation laws is identified by the law of the observations. Section 3 parametrizes the answer exactly for the shared-latent two-view design, solves it in closed form for two latent states, and gives a certified floor and a necessary condition for $k \geq 3$ (the sharp set is open, ST-O07). The identification question presumes access to the population law of (view 1, view 2) across independent world draws; an observer confined to one draw sees only $K[M, \cdot]$ for its own M. Section 4 makes this quantitative: evidence about world-level weights accrues per world draw, not per observation.

Randomness. A decision maker facing an unknown world can lower its worst-case loss with a coin that carries no information about the world. Such gains are coordination value, not learning, but they are real, and the ledger had left their cost open. Section 5 answers the direct question: what does exact shared randomness cost, in fair bits, when the optimal mixture is not dyadic?

The two halves meet in one question that this paper does not answer (Section 7): the fair-bit cost of reaching a target value in a game played against a whole identified set rather than a declared finite list of scenarios.

Contributions

Each result is listed with its claim ID; Section 2 explains the types.

1. **The two-view fiber** (ST-T297–ST-T305, ST-M86–ST-M89, ST-F59). An orthogonal parametrization of all models sharing a two-view law; local non-identification for every strictly positive channel; the sharp closed-form identified interval for $k = 2$; a certified positive floor on every latent weight; product-gauge rigidity for different channels; a necessary condition for point identification; and exact witnesses, including a no-anchor channel that two views still identify.
2. **The persistent-latent cap** (ST-T277–ST-T280, ST-F55, with ST-T273 for contrast). Expected evidence about world-level weights is at most $\mathrm{KL}(a \parallel b)$ per world draw (data processing, standard); the shortfall is an exact posterior-disagreement term with a zero-gap characterization; a new bound controls the shortfall by the MAP error.
3. **The price of shared randomness** (ST-T310–ST-T313, ST-M90–ST-M93, ST-F61). An (elementary) dyadic dichotomy, a two-sided $\Theta\big(2^{-b}\big)$ gap with no nondegeneracy assumption, the composition of the Knuth-Yao sampler with the optimal set, the analogue for L-sided coins, and exact curves and prices for the ledger's committed games.

Appendix A records how these results re-read earlier ledger entries.[1]

2 Two-view setting, notation and claim types

Persistent latent, channel, views. There are k latent states with prior π in the open simplex, and a row-stochastic $k \times d$ channel K of full row rank k. One latent $M \sim \pi$ is drawn once and emits observations that are conditionally i.i.d. given M with law $K[M, \cdot]$. Write $D_v = \mathrm{diag}(v)$. The one- and two-view laws are

$$q = K^\top \pi, \qquad T = K^\top D_\pi K \quad \text{(ST-T259)}. \tag{1}$$

Put $s = \sqrt{\pi}$ entrywise and $W = D_s K$. Then $T = W^\top W$, $W\mathbb{1} = s$, $|s| = 1$, and for the column w_x of W: $s \cdot w_x = q_x$ and $|w_x|^2 = T_{xx}$. The *two-view fiber* of (π, K) is the set of k-state models (π', K') with $K'^\top D_{\pi'} K' = T$. Throughout, k is fixed and known; alternatives with more latent states are not classified. Latents are indexed $1, \ldots, k$ and outcomes $1, \ldots, d$ (the repository indexes outcomes from 0).

Notation. The letter π always denotes a prior; the circle constant appears only inside constants, written upright ($\ln(\pi/2)$ in (8)), and angles are in degrees. T is always the two-view matrix: transcript lengths and trial counts are n, and stopping times (numbers of fair bits consumed) are τ. In Section 5 the game matrix is Γ, and K_3 names the complete confusion graph on three source states, not a channel.

Claim types. The ledger types every claim. A *theorem* (`ST-T`) follows from its stated assumptions. A *model result* (`ST-M`) is exact inside one declared model or example. A *finite check* (`ST-F`) is a reproducible computation on an explicitly bounded domain. An *open problem* (`ST-O`) is a target, not

[1] The same repository merge also added Bell-scenario results and the fresh-draw half of the evidence dichotomy (ST-T272–ST-T276). They are outside the scope of this paper, except that Section 4 uses ST-T273 as a contrast.

a conclusion. Results that are classical carry the label *standard* in the ledger, and results from the literature that are used but not reproved are *cited*. In this paper, blue blocks are the repository's own claims (theorems, model results, finite checks and open problems), and amber blocks are standard or cited results; the block header names the type and then the claim ID or the source. Proofs are given in full or sketched, with pointers to `docs/` in the repository [2], where every step is written out; each result has a module in `src/simtheory/`, a test file, and a committed receipt that a test compares byte for byte.

What is not claimed. No result here is evidence for or against simulation. "Latent = simulated versus base" is an interpretation of a mixture model, not a finding. Where a result is known, the ledger and this paper say so, and novelty is claimed only for the stated forms, with no claim of priority.

3 The two-view orthogonal fiber

3.1 Classification

Theorem (ST-T297) *Fiber classification.* Let $\pi > 0$ and let K be row-stochastic of full row rank k.

1. (π', K') is in the two-view fiber iff $K' = AK$ for a unique invertible A with $A\mathbb{1} = \mathbb{1}$, $A^\top D_{\pi'} A = D_\pi$ and $AK \geq 0$; then $\pi' = A^{-\top}\pi$.
2. Let $G(W) = \{O \in O(k) : OW \geq 0 \text{ entrywise}\}$. The map
$$O \mapsto (\pi', K'), \quad u = Os, \quad \pi' = u \odot u, \quad K' = D_u^{-1} OW \tag{2}$$
is a bijection from $G(W)$ onto the fiber, and $u > 0$ holds automatically. Two elements O, O' give models that differ only by relabeling iff $O' = PO$ for a permutation matrix P.
3. Equivalently, the fiber is in bijection with the ordered orthonormal frames contained in the dual cone $C^* = \{o : o \cdot w_x \geq 0 \text{ for all } x\}$ of the column cone $C = \text{cone}\{w_x\}$.

Proof sketch. Since $T = W^\top W$ has rank k and column space $\text{row}(K)$, any fiber point has $\pi' > 0$ and $\text{row}(K') = \text{row}(K)$, which gives $K' = AK$; cancelling a right inverse of K in $K^\top(A^\top D_{\pi'} A - D_\pi)K = 0$ gives (1). For (2), if $OW \geq 0$ then $u = OW\mathbb{1} \geq 0$, and $u_i = 0$ would make row i of OW vanish, impossible for invertible O and full-rank W. Then $A = D_u^{-1} O D_s$ satisfies $A\mathbb{1} = \mathbb{1}$ and $A^\top D_u^2 A = D_s O^\top O D_s = D_\pi$. Conversely $O = D_{\sqrt{\pi'}} A D_s^{-1}$ is orthogonal exactly when $A^\top D_{\pi'} A = D_\pi$. Part (3) restates $o_i \cdot w_x \geq 0$. Full proof: `docs/two-view-orthogonal-fiber.md`, Theorem A. □

Credit. Part (1) is already in the repository (`docs/latent-mixture-two-view-rigidity.md`, Section 6), and that two rank-k factorizations $T = W^\top W$ differ by an orthogonal matrix is the standard symmetric-NMF fact [4], [5]. The new content is the stochastic reparametrization $u = Os$, $\pi' = u \odot u$, $K' = D_u^{-1} OW$, which turns statements about the fiber into statements about orthonormal frames in a polyhedral cone.

Corollary (ST-T298) *Strictly positive channels are not even locally identified.* If $K > 0$ entrywise and $k \geq 2$, then $G(W)$ contains a neighborhood of I in $O(k)$, so the fiber contains a $k(k-1)/2$-dimensional manifold of models that are not relabelings of each other, and the set of consistent priors $\{(Os)^2 : O \in G(W)\}$ contains an open neighborhood of π in the simplex.

Proof sketch. $OW > 0$ is an open condition that holds at $O = I$. The orbit map $\text{SO}(k) \to S^{k-1}$, $O \mapsto Os$, is a submersion, hence open, and squaring is a local homeomorphism near $s > 0$. □

Corollary (ST-T299) *What the earlier rigidity theorem excludes.* $A = D_u^{-1} O D_s \geq 0$ iff $O \geq 0$, and a nonnegative orthogonal matrix is a permutation. Hence the nonnegative-gauge rigidity theorem of the ledger (ST-T261; P1-T7 of the companion paper [6]) is exactly the statement that $G(W)$ meets the nonnegative matrices only in permutations.

So ST-T261 remains correct as scoped, but its scope excludes every non-permutation point of $G(W)$, which for a strictly positive channel is the whole manifold of ST-T298. Nonnegativity of A is sufficient, not necessary, for AK to be a stochastic channel: in the witness ST-M87 below, A has a negative entry and $K' = AK > 0$.

Theorem (ST-T300) *Different channels per view.* Let $T = K_1^\top D_\pi K_2$ with K_1, K_2 row-stochastic of full row rank k. (a) Every fiber point is (π', AK_1, BK_2) with $A\mathbb{1} = B\mathbb{1} = \mathbb{1}$, $\pi' = A^{-\top}\pi$ and $B = D_{\pi'}^{-1} A^{-\top} D_\pi$; the signed parameter set is an open subset of the affine space $\{A : A\mathbb{1} = \mathbb{1}\}$, which has dimension $k^2 - k$. (b) *Product-gauge rigidity:* if moreover $A \geq 0$ and $B \geq 0$, then $A = B = P$ for one permutation P.

Proof sketch. For (b), take $j \neq l$: $0 = (D_\pi)_{jl} = \sum_i \pi'_i A_{ij} B_{il}$ with nonnegative terms, so $A_{ij} B_{il} = 0$. Fix i and j with $A_{ij} > 0$; then row i of B is e_j, and a second positive entry in row i of A would force $B_{ij} = 0$. Invertibility makes $i \mapsto j$ a permutation. □

Credit. Part (a) and its dimension count are known: the different-channel model is the setting of Henry, Kitamura and Salanié [7], and Hall and Zhou [8] show that the bivariate two-component case is identified only up to a two-parameter family. Only (b) is claimed. The same-channel fiber is the slice $A = B$, which is why its dimension drops from $k(k-1)$ to $k(k-1)/2$.

3.2 The sharp identified interval for two latent states

For $k = 2$, drop outcomes with $q_x = 0$ and set $c_x = q_x^2 / T_{xx} \in (0, 1]$. The *extreme pair* $\{x^*, y^*\}$ minimizes $T_{xy}^2 / (T_{xx} T_{yy})$; it is the pair of columns w_x, w_y spanning the widest angle, and it is observable. In any fiber model, the latent *favored by* x^* is the latent i with the larger ratio K'_{ix^*} / K'_{iy^*} (a zero denominator reads as $+\infty$). Columns x^* and y^* are not parallel in any fiber model, so this convention labels every fiber model; we call the favored latent latent 1.

Theorem (ST-T301) *Sharp identified set, $k = 2$.* With latent 1 favored by x^*, the identified set of the prior weight π'_1 is exactly

$$[L, U] = \left[1 - \frac{q_{y^*}^2}{T_{y^*y^*}}, \quad \frac{q_{x^*}^2}{T_{x^*x^*}} \right]. \tag{3}$$

The endpoints are rational in T on each region where the extreme pair is fixed and continuous on the laws with $q > 0$. U is attained by the unique fiber model in which x^* is an anchor for latent 1, and L by the model in which y^* is an anchor for latent 2. Without a label convention the identified set of π_1 is $[L, U] \cup [1 - U, 1 - L]$. Finally $L = U$ iff each latent has an anchor outcome in the true model.

Proof sketch. Up to relabeling the fiber is $\{R_\theta : R_\theta W \geq 0\}$. If the columns have angles $\varphi_x \in [0°, 90°]$ with extremes $\varphi_{\min} < \varphi_{\max}$, a rotation keeps every column in the quadrant iff $\theta \in [-\varphi_{\min}, 90° - \varphi_{\max}]$, an arc containing 0. Along it $\pi'_1 = \cos^2(\psi + \theta)$, with ψ the angle of s, is strictly decreasing, so its image is an interval whose endpoints are $\cos^2(\psi - \varphi_{\min})$ and $1 - \cos^2(\varphi_{\max} - \psi)$. Since $\cos \angle(s, w_x) = q_x / \sqrt{T_{xx}}$, these are c_{x^*} and $1 - c_{y^*}$. At $\theta = -\varphi_{\min}$ the column x^* lies on an axis, so it is an anchor. The point case is $\varphi_{\min} = 0°$ and $\varphi_{\max} = 90°$. □

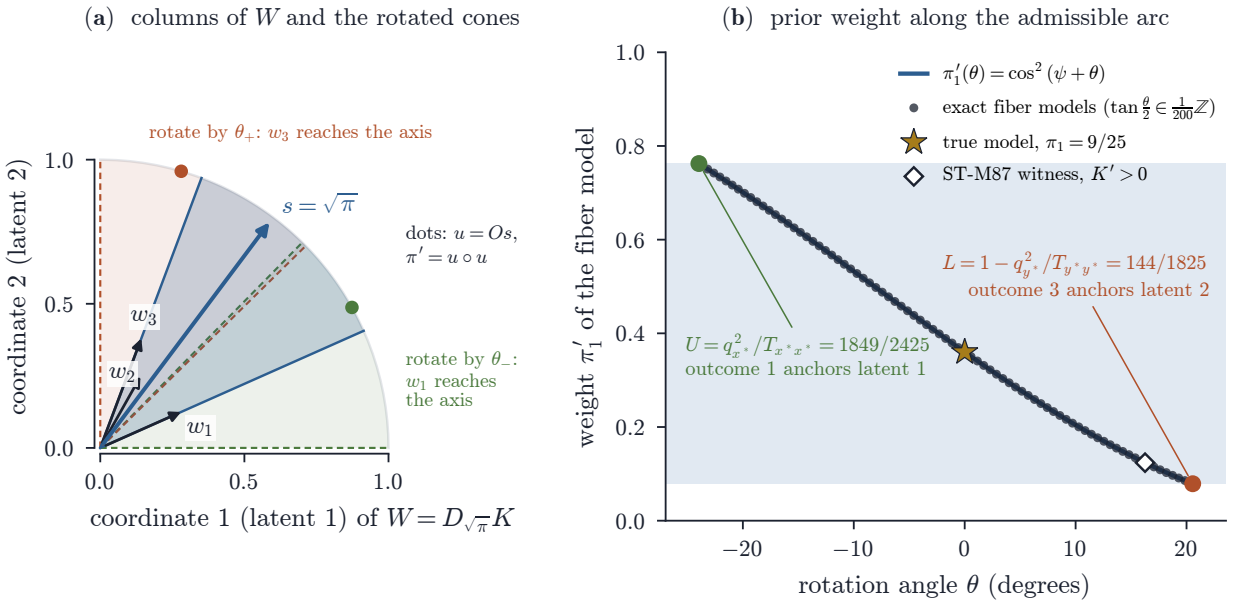


Figure 1. The $k = 2$ fiber of ST-M87. (a) The columns w_x of $W = D_{\sqrt{\pi}}K$ span the blue cone. Rotations keep all columns in the quadrant until one reaches an axis (green and rust cones); the image $u = Os$ of s moves along the unit circle (dots at the two extremes). (b) The prior weight $\pi_1' = \cos^2(\psi + \theta)$, with ψ the angle of s, along the admissible arc $\theta \in [-0.418, 0.359]$ rad (about $-24.0°$ to $20.6°$) sweeps out the identified interval $[L, U]$ (shaded). Black dots are exact fiber models on the rational grid $\tan(\theta/2) \in \{j/200 : |j| \le 100\}$, each checked in rational arithmetic to reproduce T (79 of the 201 grid points are admissible, ST-F59); the curve is the closed form. The endpoints are the anchor models.

Credit. The interval (3) specializes the sharp two-component bounds of Henry, Kitamura and Salanié [7] (their Theorem 1) to a shared channel, and "rotate until a component touches zero" is the feasible-band construction of Lawton and Sylvestre [9] in multivariate curve resolution. Only the explicit closed form for the prior weight is offered.

Model result (ST-M87) *An exact witness with a strictly positive alternative channel.* Take $\pi = \left(\frac{9}{25}, \frac{16}{25}\right)$ and $K = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}$. Then $q = \left(\frac{43}{150}, \frac{91}{300}, \frac{41}{100}\right)$, the extreme pair is $(x^*, y^*) = (1, 3)$, and

$$[L, U] = \left[\frac{144}{1825}, \frac{1849}{2425}\right] \approx [0.0789, 0.7625], \tag{4}$$

around the true 0.36. The rotation with $\cos\theta = \frac{24}{25}$, $\sin\theta = \frac{7}{25}$ gives $\pi' = \left(\frac{1936}{15625}, \frac{13689}{15625}\right)$, the signed gauge $A = \begin{pmatrix} \frac{18}{11} & -\frac{7}{11} \\ \frac{7}{39} & \frac{32}{39} \end{pmatrix}$, and the strictly positive channel $K' = \begin{pmatrix} \frac{47}{66} & \frac{13}{66} & \frac{1}{11} \\ \frac{53}{234} & \frac{149}{468} & \frac{71}{156} \end{pmatrix}$, with identical one- and two-view laws.

Figure 1 shows the geometry. The endpoint models are explicit: $K_U' = \begin{pmatrix} \frac{97}{258} & \frac{145}{516} & \frac{59}{172} \\ 0 & \frac{3}{8} & \frac{5}{8} \end{pmatrix}$ and $K_L' = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} & 0 \\ \frac{59}{246} & \frac{155}{492} & \frac{73}{164} \end{pmatrix}$. A boundary pair recorded earlier in the ledger (prior $\left(\frac{1}{5}, \frac{4}{5}\right)$ against the symmetric channel with prior $\left(\frac{1}{2}, \frac{1}{2}\right)$) is exactly the L-endpoint model of its interval $\left[\frac{1}{5}, \frac{4}{5}\right]$, and a channel with one anchor gives a one-sided interval ($\left[\frac{9}{25}, \frac{289}{325}\right]$ in ST-F59).

3.3 A certified floor for every k, and the recovery ladder

For general k the identified set is not known in closed form (ST-O07), but the cone picture yields a floor.

Theorem (ST-T302) *Weight floor.* Let $\ell(T) = \min_{o \in C^*,\ |o|\,=1} (o \cdot s)^2 = \min_\nu (\nu \cdot s)^2 / \ |\nu|^2$ over the extreme rays ν of C^*. Then $\ell(T)$ depends only on T, $0 < \ell(T) \le \min_i \pi_i$, and every model in the two-view fiber has $\pi_i' \ge \ell(T)$ for all i, hence $\pi_i' \le 1 - (k-1)\ell(T)$. For $k = 2$, $\ell(T) = \min(L, 1 - U)$ is the exact minimum latent weight.

Proof sketch. Two factors of T differ by an orthogonal Q that moves C^* and s together. The vector s is a strictly positive combination of the generators of the full-dimensional cone C, so it lies in its interior and $o \cdot s > 0$ on $C^* \setminus \{0\}$. Rows of $O \in G(W)$ are unit vectors in C^* and $\pi_i' = (o_i \cdot s)^2$. The minimum over a cone of $(o \cdot s)^2/|o|^2$ is attained on an extreme ray. □

Model result (ST-M88) *Three latent states.* For

$$\pi = \left(\frac{1}{9}, \frac{4}{9}, \frac{4}{9}\right), \quad K = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$

the floor is $\ell = \frac{4}{153} \approx 0.0261$, and it is attained by an exact orthonormal frame in C^*, so the smallest latent weight over this fiber is exactly $\frac{4}{153}$. A 9261-point rational Cayley grid contains 197 admissible points, each an exact fiber model; the different-channel witness of ST-T300 is also exact.

The grid's extreme label-free weights, $\frac{784}{23409} \approx 0.0335$ and $\frac{1369}{2025} \approx 0.676$, are attained by exact fiber models. They bound the range of the sharp set from inside but do not determine it, and a float scan suggests a largest label-free weight near 0.746 against the outer bound $1 - 2\ell \approx 0.948$ (not certified). Figure 2 shows the consistent priors in the simplex for this channel and two contrasting ones.

Theorem (ST-T303) *Two views plus anchors identify.* If each latent i has an anchor outcome x_i in the true model ($K_{jx_i} = 0$ for $j \ne i$, $K_{ix_i} > 0$), then every fiber point is a relabeling, and $\pi_i = q_{x_i}^2 / T_{x_i x_i}$. For $k = 2$ anchors are also necessary.

Proof sketch. Column x_i of AK is $A[:, i]K_{ix_i} \ge 0$, so $A \ge 0$, and ST-T299 makes A a permutation. Necessity for $k = 2$ is the point case of ST-T301. □

This is the separability (anchor-word) mechanism of Donoho and Stodden [10] and Arora, Ge, Kannan and Moitra [11]; no priority is claimed. For $k \ge 3$ anchors are not necessary:

Model result (ST-M89) *Point identification without anchors.* With π uniform on three states and

$$K = \frac{1}{8}\begin{pmatrix} 0 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 0 & 3 & 1 \\ 3 & 1 & 3 & 1 & 0 & 0 \end{pmatrix}$$

no outcome is an anchor, yet two views identify (π, K) up to relabeling. The extreme rays of C^* are e_1, e_2, e_3 and the three permutations of $(1, 1, -1/3)$; each satisfies $y \cdot \mathbb{1} \ge |y|$ with equality only on the coordinate rays, and an orthonormal frame in C^* must then consist of coordinate vectors.

This is an instance of the sufficiently scattered uniqueness theorem for symmetric NMF [4], [12]; point identification from two views is equivalent to essential uniqueness of the symmetric factorization $T = W^\top W$ with the stochastic normalization. A necessary condition holds for all k.

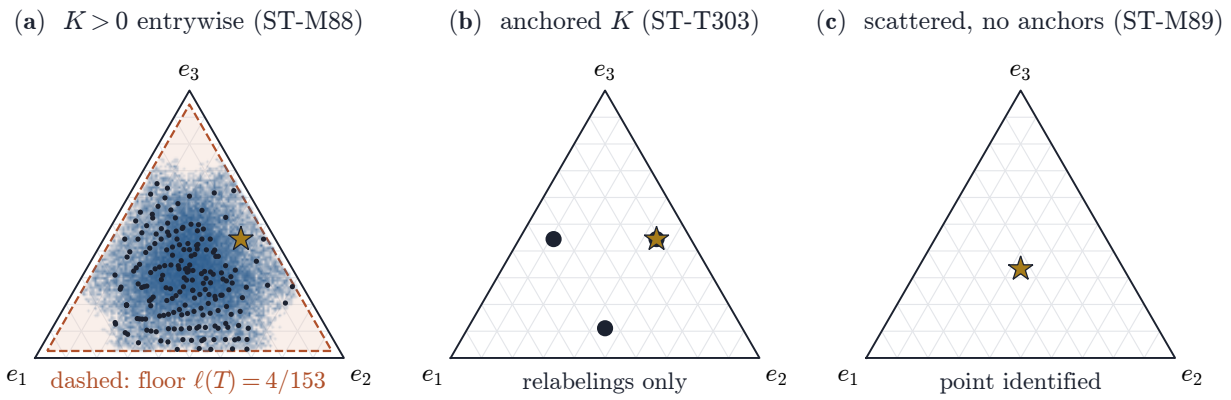


Figure 2. Consistent priors for three latent states, in the simplex with vertices e_i (star: true prior). (a) Strictly positive channel of ST-M88: a compact, label-symmetric region with nonempty interior, containing a neighborhood of π (ST-T298). Blue: a Haar float scan of $O(3)$ (3×10^6 draws, 27,064 admissible), shown for illustration only and not a certificate. Black: the 197 exact fiber models of the rational Cayley grid in ST-F59. Dashed: the certified floor $\pi'_i \geq \ell(T) = 4/153$ (ST-T302). (b) The anchored channel K with rows $(1/2, 0, 0, 1/2)$, $(0, 2/3, 0, 1/3)$ and $(0, 0, 3/4, 1/4)$, with the same prior (outcomes 1 to 3 are anchors): only relabelings survive (ST-T303), and a float scan of 2×10^5 Haar draws finds no admissible point, as expected for a finite fiber. (c) The scattered channel of ST-M89 with uniform prior: point identified without anchors.

Theorem (ST-T305) *A necessary condition for point identification.* If two views point-identify (π, K) up to relabeling, then no row support of K is contained in another: there are no $i \neq j$ with $\operatorname{supp}(K_j) \subseteq \operatorname{supp}(K_i)$. For $k = 2$ this is exactly the two-anchor condition.

Proof sketch. If $\operatorname{supp}(K_j) \subseteq \operatorname{supp}(K_i)$, rotate rows i, j of W by a small $\theta > 0$: row j becomes $\sin\theta W_i + \cos\theta W_j \geq 0$, and row i becomes $\cos\theta W_i - \sin\theta W_j$, which is nonnegative once $\tan\theta \leq \min_{x: W_{jx} > 0} W_{ix}/W_{jx}$ (positive by the support inclusion). These fiber points converge to the true model, are distinct from it, and change u_i, so they are not relabelings. □

The third rung of the ladder is cited, not reproved.

Cited theorem (Kruskal [13]) *Three views identify.* The three-view tensor $\sum_i \pi_i K_i \otimes K_i \otimes K_i$ has factor matrices $D_\pi K, K, K$ of Kruskal rank k, and $3k \geq 2k + 2$ for $k \geq 2$, so the rank-k decomposition is unique up to permutation and scaling; row-stochasticity fixes the scaling. See also [14], [15].

Model result (ST-M86) *The third view separates the witness pair.* In the $k = 2$ witness of ST-M87, the three-view laws of the true and alternative models already differ at entry $(1, 1, 1)$: $\frac{259}{5400}$ against $\frac{42413}{772200}$, with $\mathrm{KL}(P_{\text{true}} \,\|\, P_{\text{alt}}) \approx 0.00293$ nats per triple.

Table 1 collects the rungs of the ladder.

3.4 No learning inside the fiber

Standard result (ST-T304) *Bayes factor one inside the fiber.* For n i.i.d. two-view pairs the likelihood $\prod_t T[y_{1t}, y_{2t}]$ depends on the model only through T. Hence any two models in one fiber have Bayes factor one on every data set, and for any prior on k-state models the posterior given T equals the prior given T.

Table 1. The identification ladder for one persistent latent observed repeatedly through a common channel.

Data	What is identified	Claims
one view q	a row-stochastic gauge family of dimension $k^2 - k$	ST-T254–ST-T258
two views, $K > 0$	only up to $G(W) \subset O(k)$, a $k(k-1)/2$-dimensional set; prior not locally identified	ST-T297, ST-T298
two views, $k = 2$	the interval (3); a point iff both latents have anchors	ST-T301, ST-T303
two views, any k	floor $\ell(T)$; point identification needs incomparable supports	ST-T302, ST-T305
two views + anchors, or sufficiently scattered K	point (up to labels)	ST-T303; cited [4] (instance ST-M89)
three views	point (up to labels), by Kruskal's theorem	cited [13]

This is the standard "no learning about non-identified directions" fact [16], [17]; only its location inside the two-view fiber is new. It is the ledger's boundary ST-T01 reappearing inside a restricted, otherwise informative model. A cited remark completes the picture: under the regularity conditions of Giacomini and Kitagawa [18] (a consistent reduced-form posterior and continuity of the identified-set correspondence at the true T), the set of posterior means of the label-fixed weight over all conditional priors on the fiber converges to $[L, U]$, not to a point; Moon and Schorfheide [19] give the related single-prior statement. The continuity condition is not automatic. With $\pi = \left(\frac{1}{2}, \frac{1}{2}\right)$ and K_ε with rows $(3/4 - \varepsilon, 1/4, \varepsilon)$ and $(1/4, 3/4, 0)$, $U = \frac{1}{2}$ for every $\varepsilon > 0$ (outcome 3 is an anchor) while $U = \frac{4}{5}$ at $\varepsilon = 0$; ST-F59 confirms this exactly.

Finite check (ST-F59) *Exact fiber audit.* In rational arithmetic: the $k = 2$ witness and its converse; a 201-point rational grid around the arc (79 admissible), with probes at distance 10^{-6} inside and outside both endpoints; anchored $k = 3$ (none of 728 non-identity Cayley points admissible); the 9261-point $k = 3$ grid and the exact floor certificate; the different-channel grids (49 of 49 signed gauges admissible; only the two common permutations survive among 25 nonnegative product gauges); the scattered channel (0 of 2196 non-identity points); the nested-support rotation; and the jump of U.

4 The persistent-latent cap: the evidential counterpart

Section 3 asked which hypotheses the observable law can separate at all. This section asks how much evidence can accumulate when the hypotheses differ only in how likely each world is. Let M take values in a standard Borel space, let a, b be two laws of M ("world-level weights"), and let a Markov kernel $K(dy \mid m)$ generate the whole transcript Y, which may have any length, be randomly stopped, or be adaptive under a policy that does not depend on the hypothesis label. Write $P_a^Y = aK$ and $P_b^Y = bK$.

Standard result (ST-T277) *Persistent cap.*

$$\mathrm{KL}(aK \parallel bK) \le \mathrm{KL}(a \parallel b), \tag{5}$$

and with N independent world draws $M_1, \ldots, M_N$, i.i.d. from a (resp. b), and any common kernel from $(M_1, \ldots, M_N)$ to Y, $\mathrm{KL}(P_a^Y \parallel P_b^Y) \le \mathrm{KL}(a^{\otimes N} \parallel b^{\otimes N}) = N\,\mathrm{KL}(a \parallel b)$, with equality to $\sum_u \mathrm{KL}(aK_u \parallel bK_u)$ when each unit's transcript depends only on its own draw through a kernel K_u and the units are independent.

This is the data-processing inequality for KL divergence [20], [21], not claimed as new; the contribution is its application to world-level weights. Its meaning for the simulation question is direct: expected

evidence about how likely each kind of world is scales with the number of independent world draws, not with transcript length. It is the expected (KL) form of the pointwise ceiling $\ln \max_m a_m/b_m$ of ST-T226, and it does not limit evidence about *which* world is active. When a is not absolutely continuous with respect to b, (5) is vacuous, and the two sides need not be infinite together: $a = \left(\frac{1}{2}, \frac{1}{2}\right)$, $b = (1, 0)$ with identical component laws give $\mathrm{KL}(a \parallel b) = +\infty$ but transcript divergence 0 for every length.

Theorem (ST-T278) *Saturation identity.* If $\mathrm{KL}(a \parallel b) < \infty$, the *saturation gap* $\mathrm{gap} := \mathrm{KL}(a \parallel b) - \mathrm{KL}(aK \parallel bK)$ satisfies

$$\mathrm{gap} = \mathbb{E}_{Y \sim aK}[\mathrm{KL}(a(\cdot \mid Y) \parallel b(\cdot \mid Y))]. \tag{6}$$

For finite M with component densities k_m (with respect to a common dominating measure), put $r_m = a_m/b_m$ and $S_b(y) = \{m : b_m k_m(y) > 0\}$. The gap is zero iff r is constant on $S_b(y)$ for aK-almost every y. In particular the gap is zero if M is a function of Y; the converse holds when the ratios r_m are pairwise distinct on $\mathrm{supp}(b)$ and fails in general.

The identity is the KL chain rule, and the zero-gap condition specializes the Kullback-Leibler equality condition for data processing [22]; only the finite-M ratio form is stated here. The converse fails: with $a = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right)$, $b = \left(\frac{1}{8}, \frac{1}{8}, \frac{3}{4}\right)$, two components sharing the law $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$ and a third with law $(0, 0, 1)$, the transcript divergence equals $\mathrm{KL}(a \parallel b) \approx 0.1438$ at lengths 1, 3 and 6, yet M is not a function of Y (ST-F55 records a second, deterministic counterexample). The main new piece of this section is a quantitative bound on the gap.

Theorem (ST-T279) *Gap bound and decay.* Let M be finite with $a \ll b$, $S = \mathrm{supp}(a)$, $Z = \mathrm{supp}(b) \setminus S$, $r_{\max} = \max_{m \in S} r_m$, $r_{\min} = \min_{m \in S} r_m$, and let $P_a(\mathrm{err})$ be the Bayes error of the MAP rule for M under a. For any orders $\lambda_{mm'} \in (0, 1]$,

$$\mathrm{gap} \le \left(\frac{r_{\max}}{r_{\min}} - 1\right) P_a(\mathrm{err}) + \sum_{m \in S, m' \in Z} \frac{a_m}{\lambda_{mm'}} \left(\frac{b_{m'}}{b_m}\right)^{\lambda_{mm'}} B_{\lambda_{mm'}}(k_m, k_{m'}), \tag{7}$$

where $B_\lambda(P, Q) = \int p^{1-\lambda} q^\lambda d\mu$. For conditionally i.i.d. transcripts of length n whose relevant component pairs have distinct laws, the gap therefore decays at least as fast as $\exp(-n(\min C(k_m, k_{m'}) - o(1)))$, the minimum of Chernoff informations over relevant pairs (pairs within S when $r_{\max} > r_{\min}$, and pairs in $S \times Z$).

Proof sketch. Fix y, let α and β be the posteriors of M under a and b, and let $p_b^S(y) = \sum_{m \in S} b_m k_m(y)$ and $p_b^Z(y) = \sum_{m \in Z} b_m k_m(y)$ be the parts of the b-mixture density from S and from Z. Then $\mathrm{KL}(\alpha \parallel \beta) = \{\mathbb{E}_\alpha \ln r + \ln \mathbb{E}_\alpha[1/r]\} + \ln\left(1 + p_b^Z/p_b^S\right)$. With $\varepsilon = 1 - \max \alpha$ the bracket is at most $\varepsilon(r_{\max}/r_{\min} - 1)$ by convexity, and its expectation is $(r_{\max}/r_{\min} - 1)P_a(\mathrm{err})$. The last term is bounded with $\ln(1 + x) \le x^\lambda/\lambda$. The rate follows from the classical multi-hypothesis Chernoff bound on the MAP error [23], [24]. Full proof: `docs/evidence-scaling-dichotomy.md`. □

The MAP-error exponent is classical; only the $(r_{\max}/r_{\min} - 1)$ factor and the support-mismatch term are new, and the rate is an upper bound on the exponent, not shown to be tight. Distinct component laws are needed. Take binary transcripts with success probabilities $\eta_m = P(Y_t = 1 \mid M = m)$ equal to $\left(\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right)$, $a = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ and $b = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$: two components coincide, the transcript divergence never exceeds the merged value 0.2426, and the gap stays at least 0.1236 for every n.

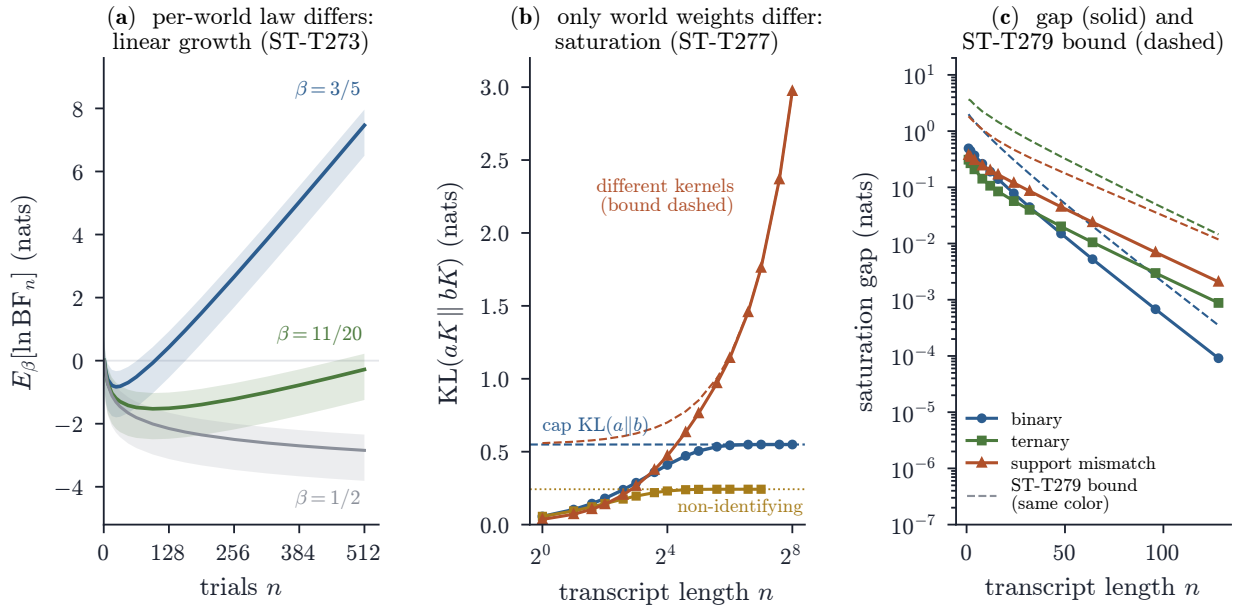


Figure 3. Two regimes of evidence. (a) Per-world law differs: $\mathbb{E}_\beta[\ln \mathrm{BF}_n]$ for the Krichevsky-Trofimov Bayes factor against fair bits, computed exactly (60-digit arithmetic), with the ST-T273 sandwich (8) shaded for each β; at $\beta = 1/2$ it decreases like $-\frac{1}{2}\ln n$. (b) Only world weights differ: transcript divergence for one persistent binary latent ($\eta = (1/3, 2/3)$, $a = (3/4, 1/4)$, $b = (1/4, 3/4)$) saturates at $\mathrm{KL}(a \parallel b) = 0.5493$ (ST-T277); with the non-identifying components of Section 4 (gold) it saturates at the merged value 0.2426; with different kernels it passes the cap and tracks the ST-T280 bound (dashed), within 10^{-6} of it at $n = 256$. (c) The saturation gap (solid) and the ST-T279 bound (dashed, same color) on three ST-F55 examples. The bound is valid at every n. For the binary and ternary examples it uses the exact MAP error; for the support-mismatch example its mismatch term is evaluated at $\lambda = 1/2$ (Bhattacharyya), whose exponent 0.034668 is within 0.1% of the Chernoff exponent 0.034688.

Standard result (ST-T280) *Different kernels.* If the hypotheses also use different kernels K^a, K^b, then $\mathrm{KL}\big(aK^a \parallel bK^b\big) \le \mathrm{KL}(a \parallel b) + \int \mathrm{KL}\big(K^a(\cdot \mid m) \parallel K^b(\cdot \mid m)\big)a(dm)$, and for conditionally i.i.d. transcripts the second term is linear in n.

The dichotomy. The contrast with hypotheses that disagree about the per-world law is sharp. For n tosses of a coin of bias β, with fair bits as the null and a Krichevsky-Trofimov (Jeffreys) mixture of biased coins as the alternative, the ledger proves the two-sided bound (ST-T273)

$$n\,\mathrm{KL}\Big(\beta \parallel \frac{1}{2}\Big) - \frac{1}{2}\ln n - \ln 2 \le \mathbb{E}_\beta[\ln \mathrm{BF}_n] \le n\,\mathrm{KL}\Big(\beta \parallel \frac{1}{2}\Big) - \frac{1}{2}\ln n + 1 - \frac{1}{2}\ln\Big(\frac{\pi}{2}\Big) \tag{8}$$

for every $n \ge 1$ and $\beta \in [0, 1]$, with exact constant $\frac{1}{2}\ln(2e/\pi)$ for interior β (ST-T274, the Clarke-Barron constant [25] specialized; the upper regret bound is Krichevsky-Trofimov [26]). The alternative in (8) is itself one persistent draw of the bias, so the dividing line is not "fresh versus persistent". It is whether the hypotheses differ in the per-world law ($\mathrm{KL}(a \parallel b) = +\infty$ in the cap, or different kernels as in ST-T280) or only in world-level weights over a common kernel. Figure 3 shows both regimes on the ledger's exact examples.

Finite check (ST-F55) *Exact persistent-latent audit.* Exact laws over sufficient count vectors with 60-digit logarithms, for binary, ternary, support-mismatch, ternary-emission and non-identifying examples up to $n = 64$: monotonicity, the cap, the identity of ST-T278 against an independent computation to within 10^{-30}, the bound of ST-T279 and its Bhattacharyya relaxation, the two counterexamples, and additivity over two independent world draws.

Relation to Section 3. The two sections describe one structure from two sides. Inside a two-view fiber the evidence is exactly zero (ST-T304). Outside it, if two hypotheses share the channel and differ only in the prior over latent states, then any transcript of a single draw, including the two-view and three-view transcripts of Section 3, gives at most $\mathrm{KL}(a \parallel b)$ nats of expected evidence about the prior (ST-T277 applies to every transcript design). Evidence beyond that per-draw budget requires either more independent world draws or hypotheses that disagree about the per-world laws.

5 The price of shared randomness

5.1 Setting

A finite rational game $\Gamma \in \mathbb{Q}^{R \times J}$ has rows r for adversary scenarios (point-mass models, declared priors, or vertices of a total-variation ball) and columns $c = 1, \ldots, J$ for deterministic zero-error codebooks; Γ_{rc} is the expected codeword length, or regret, of codebook c under scenario r. Write Δ for the simplex of column laws w.

The K_3 game. Three source states are pairwise confusable (the complete confusion graph K_3), so a zero-error code gives them distinct codewords. The scenarios are the three point masses, and the columns are the three complete prefix codes whose codeword lengths are a permutation of $(1, 2, 2)$: $\Gamma_{mc} = 1$ if code c gives state m the one-bit codeword, and 2 otherwise. Every deterministic code has worst case 2, while the uniform mixture has worst case $5/3$. This game recurs in the ledger (ST-M26, ST-M41, ST-M45, ST-M47) and is the running example below.

The assumptions are those of `docs/random-bit-price-shared-randomness.md`, Section 1:

(A1) Public seed. A seed is observed by the encoder and every decoder before the source is observed, and it selects a column. Every column is zero error, so zero error is preserved.

(A2) Oblivious adversary. The scenario is chosen without seeing the seed. A seed that induces column law $w \in \Delta$ therefore has payoff $f(w) = \max_r (\Gamma w)_r$, and $V^* = \min_{w \in \Delta} f(w)$, with optimal set $\mathcal{O}$, a nonempty rational polytope.

(A3) Resource models. Fixed budget: b independent fair bits and any deterministic map from them to columns (value V_b). Variable budget: an i.i.d. fair bit stream, an almost surely finite stopping time τ of its natural filtration, and an output that is a function of the bits read. L-ary coin: t independent uniform L-sided tosses and a deterministic map (value $V_{L,t}$).

(A4) Rationality. The entries of Γ are rational with common denominator D_Γ, and $V^* = p/Q$ in lowest terms.

Standard result (elementary) *Realizable laws.* With b fair bits the realizable column laws are exactly $\Delta_b = \Delta \cap 2^{-b}\mathbb{Z}^J$; with t uniform L-ary tosses, exactly $\Delta \cap L^{-t}\mathbb{Z}^J$.

We refer to this as Lemma 0, as the repository does.

5.2 Fixed budgets: a dichotomy and a two-sided rate

Theorem (ST-T310) *Dyadic dichotomy.* $V^* \le V_{b+1} \le V_b$, and $V_b = V^*$ iff $\mathcal{O}$ meets Δ_b. Hence V^* is reached by finitely many fair bits iff some optimal mixture is dyadic, and then for all $b \ge b_0$.

Theorem (ST-T311) *Two-sided gap without nondegeneracy assumptions.* For every optimal w^* with support S, $\sigma = |S|$ and payoff spread $\delta_S = \max_r(\max_{c\in S}\Gamma_{rc} - \min_{c\in S}\Gamma_{rc})$,

$$V_b - V^* \le \delta_S \Big\lfloor \frac{\sigma}{2} \Big\rfloor 2^{-b}, \tag{9}$$

and whenever $V_b > V^*$, $V_b - V^* \ge 1/\operatorname{lcm}\big(2^b D_\Gamma, Q\big) \ge 2^{-b}/(D_\Gamma Q)$. So if $\mathcal{O}$ contains no dyadic point,

$$\frac{2^{-b}}{D_\Gamma Q} \le V_b - V^* \le \delta_S \Big\lfloor \frac{\sigma}{2} \Big\rfloor 2^{-b} \quad \text{for every } b \ge 0. \tag{10}$$

Proof sketch. Upper bound: round Nw^* ($N = 2^b$) by largest remainders (Hamilton apportionment) inside S; the positive part of the error e is at most $\lfloor\sigma/2\rfloor/N$, and $(\Gamma e)_r \le \delta_S$ times it in every row. Lower bound: V_b lies in $\big(2^b D_\Gamma\big)^{-1}\mathbb{Z}$ and $V^* = p/Q$, so a positive difference is at least $1/\operatorname{lcm}\big(2^b D_\Gamma, Q\big)$. □

The lower constant is attained at even b by the K_3 game and by the skew total-variation game of ST-M91 (Section 5.4); the upper constant is not claimed tight. By the Shapley-Snow basic-solution theorem [27] there is an optimal w^* with $\sigma \le \min(R, J)$, which gives a bound in terms of the payoff range. The classical sparse-strategy results of Althöfer [28] and Lipton and Young [29] give N-uniform ε-optimal strategies with $N = O(\log R/\varepsilon^2)$, a gap of order $\sqrt{\ln R/2^b}$ that is worse in b but independent of σ and J; see also [30], [31] on playing with limited randomness. The new content of (10) is the exact $\Theta\big(2^{-b}\big)$ rate for a fixed rational game with no nondegeneracy assumption. Deciding whether $\mathcal{O}$ contains a dyadic point is exact: for $L \ge 2$, a rational polytope meets $\mathbb{Z}[1/L]^J$ iff its affine hull does (Lemma 2a of `docs/random-bit-price-shared-randomness.md`), which reduces to a finite congruence check.

5.3 Variable budgets: the Knuth-Yao price

For $x \in [0,1]$ let $\kappa(x) = \sum_{t\ge 0} \operatorname{frac}(2^t x)2^{-t}$, so $\kappa(1) = 0$; for $x \in [0,1)$ with binary digits x_j (the terminating expansion when x is dyadic), $\kappa(x) = \sum_{j\ge 1} j x_j 2^{-j}$. For a law w let $\rho_t(w) = \sum_c \operatorname{frac}(2^t w_c)2^{-t}$.

Standard result (Knuth-Yao [32]; reproved in the repository) *Optimal exact sampling.* Every exact sampler of w from fair bits satisfies $P(\tau > t) \ge \rho_t(w)$ for all t. The discrete distribution generating (DDG) tree attains this bound at every t, so it minimizes $\mathbb{E}[\tau]$, with $\mathbb{E}^*[\tau](w) = \sum_c \kappa(w_c)$, a rational number when w is rational. Moreover $H(w) \le \mathbb{E}^*[\tau](w) < H(w) + 2$ [20].

The repository reproves the first two statements; the upper bound $H + 2$ is cited and checked numerically on 200 seeded random laws.

Theorem (ST-T312) *The price of the value.* V^* is attainable with an almost surely finite number of fair bits in every finite rational game. The infimum of the expected number of fair bits over samplers attaining V^* equals $\min_{w\in\mathcal{O}} \sum_c \kappa(w_c)$, and the minimum is attained. If $\tau \le B$ almost surely then the induced law lies in Δ_B; so if $\mathcal{O}$ has no dyadic point, no bounded-bit sampler attains V^*, and every sampler that does has $P(\tau > t) \ge \min_{w\in\mathcal{O}} \rho_t(w) > 0$ for every t.

Proof sketch. Attaining V^* means inducing a law in $\mathcal{O}$; apply the Knuth-Yao statements and Lemma 0. The functions κ and ρ_t are lower semicontinuous and $\mathcal{O}$ is compact. □

Only the composition with the optimal set is claimed; when $\mathcal{O}$ is not a singleton the minimum is attained but not computed in general.

Theorem (ST-T313) *Coins with L sides.* For integers $L \geq 2$, $V_{L,t} = V^*$ iff $\mathcal{O}$ meets $L^{-t}\mathbb{Z}^J$; V^* is reached by finitely many L-ary tosses iff $\mathcal{O}$ meets $\mathbb{Z}[1/L]^J$; and whenever $V_{L,t} > V^*$, $V_{L,t} - V^* \geq 1/(D_\Gamma Q L^t)$.

Building a uniform L-ary coin from fair bits is itself priced by the Knuth-Yao theorem and is not free.

5.4 Worked instances

Model result (ST-M90) *The K_3 value curve.* In the K_3 game,

$$V_b = 2 - \frac{\lfloor 2^b/3 \rfloor}{2^b}, \quad V_b - \frac{5}{3} = \frac{2^b \bmod 3}{3 \cdot 2^b} > 0, \quad V_{2j} = V_{2j+1} = \frac{5}{3} + \frac{1}{3 \cdot 4^j}. \tag{11}$$

The sequence is $2, 2, \frac{7}{4}, \frac{7}{4}, \frac{27}{16}, \frac{27}{16}, \frac{107}{64}, \ldots$: $V^* = \frac{5}{3}$ is never reached, every odd-numbered bit is worthless, and for $j \geq 1$ the $2j$-th bit is worth exactly 4^{-j}.

Proof sketch. $(\Gamma w)_m = 2 - w_m$, so $f(w) = 2 - \min_m w_m$; on Δ_b the best minimum is $\lfloor 2^b/3 \rfloor / 2^b$, and $2^b \bmod 3$ alternates between 1 and 2. □

Model result (ST-M91) *The skew total-variation game at radius one half.* For the K_3 graph with nominal prior $\left(\frac{4}{5}, \frac{1}{10}, \frac{1}{10}\right)$ and total-variation radius $1/2$ (the game of ST-M27, whose five rows are the vertices of the ball and whose three columns give the one-bit codeword to states 2, 1 and 3), $\mathcal{O} = \left\{\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)\right\}$, $V^* = \frac{5}{3}$, and

$$V_b = \frac{5}{3} + \frac{2^b \bmod 3}{30 \cdot 2^b}, \tag{12}$$

exactly one tenth of the K_3 gap. The gain over the deterministic value $17/10$ is $\lfloor 2^b/3 \rfloor / (10 \cdot 2^b)$, zero for $b = 0$ and $b = 1$.

For radius at most $7/15$ the deterministic code that gives the heavy state the one-bit codeword is already optimal (proved in `docs/random-bit-price-shared-randomness.md`, Section 4, and checked on the grid $r = k/30$), so below the ST-M27 threshold there is no randomization gain and nothing to pay for.

Model result (ST-M92) *Knuth-Yao prices of the committed instances.* In the reduced games (Table 2), the optimal set is the singleton uniform law for the K_3 game (ST-M26, ST-M45) and for ST-M27 at radius $1/2$, so no fixed budget reaches V^* and the least expected cost is $3\kappa(1/3) = 8/3$ fair bits. It is the singleton $(1/2, 1/2)$ law for ST-M21 (minimax and regret) and for the shared-seed game on the ST-M22 instance, so one bit suffices and is necessary. At radii $2/5$ and $7/15$ of ST-M27 it contains a pure law, so no bits are needed.

The $8/3$-bit cost of an exact uniform ternary choice is classical [32], [33]; ST-M92 identifies it as the exact coordination price of these games. The ST-M22 instance has four pairwise-confusable states and two priors that put $7/10$ on different states; its reduced shared-seed game has rows $(3/2, 2, 21/10)$ and $(21/10, 2, 3/2)$, deterministic value 2 and $V^* = 9/5$ at $(1/2, 0, 1/2)$. The two-scenario gain of ST-M21, from $19/10$ to $31/20$ in worst-prior expected length, costs exactly one fair bit, fixed or expected.

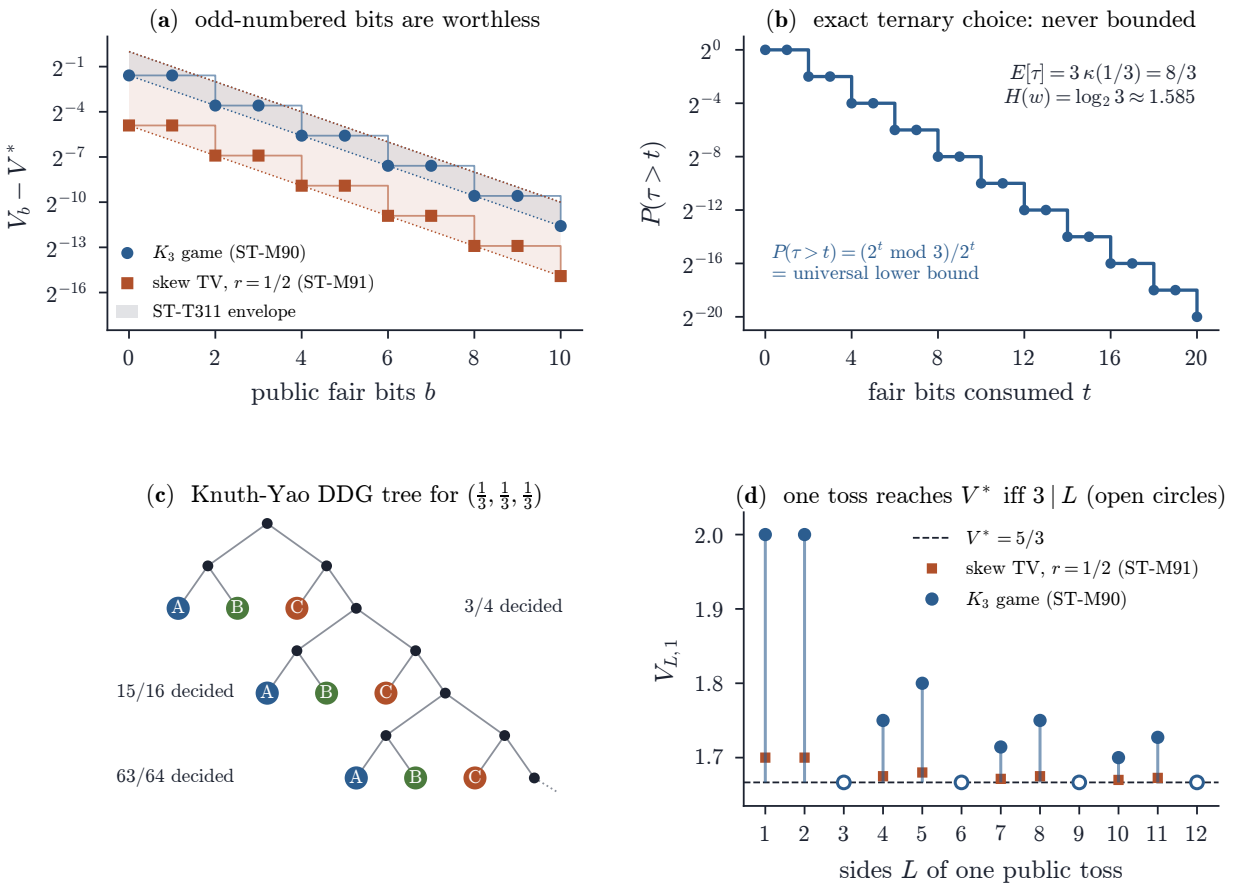


Figure 4. The fair-bit price of shared randomness. (a) Exact gaps $V_b - V^*$ for the K_3 game (ST-M90) and the skew TV game at radius 1/2 (ST-M91), each computed by exhaustive search over the 2^b-grid, with the ST-T311 envelopes shaded; the even-b values sit on the lower envelope, and every odd-numbered bit is worthless. (b) For the uniform ternary law, the Knuth-Yao tail $P(\tau > t) = (2^t \bmod 3)/2^t$ equals the universal lower bound at every t (checked to depth 40 in ST-F61, plotted to 20) and never reaches zero; $\mathbb{E}[\tau] = 8/3$. (c) The DDG tree for the uniform ternary law: leaves A, B and C are the three outcomes, and each pair of levels decides three quarters of the remaining mass. (d) One public L-sided toss reaches $V^* = 5/3$ exactly when $3 \mid L$, for both games (ST-M93). All values are exact rationals from `random_bit_price.py`.

Model result (ST-M93) *One L-sided toss.* One uniform L-ary public toss attains V^* iff $3 \mid L$ for the K_3 game and for ST-M27 at radius 1/2 (checked for $L \le 12$, and for the K_3 game also for all $L^t \le 1000$ with $3 \nmid L$), with one-toss K_3 value $V_{L,1} = 2 - \lfloor L/3 \rfloor / L$; and iff $2 \mid L$ for ST-M21 and the ST-M22 game ($L \le 12$). The general criterion is ST-T313.

Finite check (ST-F61) *Exact certification of the bit price.* The K_3 game for $b \le 12$ by exhaustive enumeration; eight committed matrices for $b \le 10$ with an independent basis-enumeration solver and dual certificates, both gap bounds and the dyadic decision; 300 seeded random integer games ($b \le 6$; 219 dyadic and exact, 81 never exact); 200 seeded random rational laws for the Knuth-Yao identities; coin alphabets $L \le 12$; and cross-checks of seven matrices against the repository's own certificates.

Table 2. Exact random-bit prices of the committed shared-randomness instances (model results ST-M92 and ST-M93, certified in ST-F61). The last row has no randomization gain. Columns: optimal law; least fixed budget b with $V_b = V^*$; least expected number of fair bits attaining V^*; when one uniform L-ary toss attains V^*. [a] In the reduced game, which merges payoff-identical columns and moves weakly dominated weight onto dominators; this changes none of the values (Lemma P of `docs/random-bit-price-shared-randomness.md`).

Instance	Optimal law[a]	Fixed b	Expected bits	L-ary toss
K_3 game (ST-M90; ST-M26, ST-M45)	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, unique	none	$3\kappa(\frac{1}{3}) = \frac{8}{3}$	iff $3 \mid L$
skew TV, $r = 1/2$ (ST-M91; ST-M27)	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, unique	none	$\frac{8}{3}$	iff $3 \mid L$
ST-M21 (minimax and regret)	$(\frac{1}{2}, \frac{1}{2})$, unique	$b = 1$	1	iff $2 \mid L$
ST-M22 instance, shared seed	$(\frac{1}{2}, 0, \frac{1}{2})$, unique	$b = 1$	1	iff $2 \mid L$
ST-M27, $r \in \{2/5, 7/15\}$	contains a pure law	$b = 0$	0	any L

6 Limitations and nonclaims

Interpretation. Nothing in this paper is evidence for or against the hypothesis that reality is simulated. The results bound what restricted, declared models allow an observer to learn or to buy; reading a latent state as "simulated versus base" is an interpretation of the mixture model, not a finding.

Scope of the fiber results. Finite alphabets, $\pi > 0$, K of full row rank k, k known, and exact population laws of the two views across independent world draws; an observer of a single draw never sees T. Alternatives with more than k latent states are not classified (splitting a state into identical copies enlarges the set of consistent weights), and alternatives with fewer are impossible because rank $T = k$. There is no finite-sample estimator or confidence interval for $[L, U]$. Nothing for $k \geq 3$ is sharp beyond the instance: the $k = 3$ numbers are exact values attained by fiber models, an exact floor attained on one instance, and an uncertified float estimate. Kruskal's three-view theorem and the Giacomini-Kitagawa limit are cited, not proved. Anchors in ST-T303 refer to the true channel. The upper bound $1 - (k-1)\ell$ is not claimed sharp.

Scope of the cap. ST-T277 and ST-T280 are standard; ST-T277 bounds an expectation, not every transcript (that is ST-T226), is vacuous when a is not absolutely continuous with respect to b, and does not limit learning which world is active. The policy must not depend on the hypothesis label. The ST-T279 exponent is an upper bound, not shown tight, and needs distinct component laws.

Scope of the bit price. Fair, independent, public bits (or uniform L-ary tosses) are assumed; extracting them from a biased or physical source is not addressed [34], and no fairness of any real source is certified. The adversary is oblivious to the seed; if the scenario may depend on the realized seed, no number of bits helps. Private encoder randomness is not covered. Payoffs must be rational. The upper constant of ST-T311 is not claimed tight; the minimum over a non-singleton $\mathcal{O}$ is attained but not computed in general; approximate sampling under a bit budget is treated only through V_b (for optimal fixed-precision approximate sampling see [35]). The congruence decision procedure is exact but bounded in the checker. Knuth-Yao's theorem and the 8/3-bit cost of a uniform ternary choice are classical, and the dyadic dichotomy ST-T310 is elementary.

Novelty. In the fiber section, part (1) of ST-T297 is in the repository already, the orthogonal form is the standard symmetric-NMF fact, part (a) of ST-T300 is known, anchor sufficiency is known, and ST-T304 is standard. The stochastic reparametrization, the closed form of the $k = 2$ interval, the floor $\ell(T)$, product-gauge rigidity, the necessary condition ST-T305 and the exact witnesses were not found stated in this form in the literature we checked. In the evidence section only the $(r_{\max}/r_{\min} - 1)$ factor and the mismatch term of ST-T279 and the finite-M ratio form of ST-T278 are offered as new. In the randomness section, the exact two-sided rate without nondegeneracy assumptions, the

composition of Knuth-Yao with the optimal set, and the exact curves of the committed instances are offered as new. No priority is claimed for any of them.

7 Open problems

Open problem (ST-O07) *Point identification and sharp sets for $k \geq 3$.* (1) Characterize point identification from two views exactly: it lies between pairwise-incomparable row supports (necessary, ST-T305) and the sufficiently scattered condition (sufficient [4], which is NP-complete to check); see also [5]. (2) Describe the sharp sets $\{(Os)^2 : O \in O(k), OW \geq 0\}$, the orthonormal frames inside a polyhedral cone; on the $k = 3$ witness the float estimate of the largest label-free weight is about 0.746 against the generic 0.948 (compare the rotational-ambiguity literature [36]). (3) Give finite-sample, uniformly valid confidence sets for $[L, U]$; the endpoints are min and max functionals, non-smooth at ties.

Open problem (ST-O02) *Second-order term of the evidence constant.* Is the remainder of ST-T274 equal to $(1-2\beta)^2/(24n\beta(1-\beta)) + O(n^{-2})$ for interior β? A formal moment expansion gives this and the numerics agree, but the dominated remainder bounds are not written, so it stays open.

Further questions are not registered as claims but are marked by nonclaims above: compute $\min_{w \in \mathcal{O}} \sum_c \kappa(w_c)$ when the optimal set is a polytope rather than a point (ST-T312 proves only that the minimum is attained); decide whether the Chernoff exponent of ST-T279 is tight, and whether the upper constant $\delta_S \lfloor \sigma/2 \rfloor$ of ST-T311 can be improved; and classify alternatives with more than k latent states. The question that joins the two halves of this paper, also unaddressed by the ledger, is the fair-bit cost of reaching a target value in a game played against a whole identified set rather than a declared finite scenario list.

Reproducibility

All results are merged on the repository's main branch: PR #72 (squash commit `0c77232`) and the attribution sweep PR #73 (`aa61193`). Clone the repository [2] at `aa61193` and run `uv sync && uv run python -m simtheory verify`. The claim manifests are `claims/two-view-orthogonal-fiber-claims.json`, `claims/evidence-scaling-dichotomy-claims.json` and `claims/random-bit-price-claims.json`; the derivations are `docs/two-view-orthogonal-fiber.md`, `docs/evidence-scaling-dichotomy.md` and `docs/random-bit-price-shared-randomness.md`; the certificates are `src/simtheory/two_view_fiber.py`, `evidence_scaling.py` and `random_bit_price.py`; and the committed receipts are `receipts/two-view-orthogonal-fiber.txt`, `receipts/evidence-scaling-dichotomy.txt` and `receipts/random-bit-price-shared-randomness.txt`. The README at that commit records 465 claims in 60 manifests. For this paper the three certificate modules were rerun from that commit and reproduced their committed receipts byte for byte. Every figure is generated by `figures.py` (next to this paper's source) from those modules. The only non-exact plotted quantities are closed-form curves (the $k = 2$ curve and its arc endpoints in Figure 1, and the ST-T273 envelope in Figure 3) and the illustrative Haar scan in Figure 2; all are labeled in the captions.

A Relation to earlier ledger entries

The fiber, the cap and the bit price re-read several earlier ledger entries without changing any of them. Pull-request numbers refer to the repository history.

A.1 Mixture results

- ST-T254–ST-T258 (one-view gauge, PR #63): the one-view gauge family $\{A : A\mathbb{1} = \mathbb{1}\}$ has dimension $k^2 - k$; the second view cuts it to $G(W)$, of dimension at most $k(k-1)/2$, full-dimensional when $K > 0$ and possibly finite otherwise (ST-M89).
- ST-T261, ST-T263 (two-view rigidity, PR #64) and P1-T7 of the companion paper [6] (PR #69): correct as scoped; ST-T299 identifies the scope as "nonnegative orthogonal matrices are permutations". ST-T263 is correct but partial: two views shrink the fiber without collapsing it. Part (b) of ST-T300 supplies the different-channel analogue that the P1-T7 nonclaims set aside.
- ST-M77–ST-M79 and ST-F47: all boundary witnesses remain correct. ST-M87 adds the first witness whose alternative channel is strictly positive; the existing boundary pair (PR #64 and the PR #68 audit) is an anchor (endpoint) model.
- ST-T225–ST-T228, ST-M66, ST-M67 (persistent-latent ceilings, PR #56): ST-T277 is the expected (KL) form of the pointwise ceiling ST-T226; ST-T278 and ST-T279 quantify the saturation of ST-T227; the N-unit form is the KL form of ST-T228 and, with ST-T233, says evidence about mixing weights scales with independent world draws. ST-T280 complements the ST-T226 scope nonclaim about component distinguishability.

A.2 Coding results

Read through ST-M90, the earlier K_3 entries acquire exact finite-budget versions. The minimax regret of ST-M45 with b bits is $1 - \lfloor 2^b/3 \rfloor / 2^b$, never $2/3$. The coordination term $1/3$ of ST-M47 is realized as $\lfloor 2^b/3 \rfloor / 2^b$, so one bit buys nothing. ST-M41's public-randomness game is, by its declared scope (three point-mass K_3 models and a source-independent public signal), the K_3 game of ST-M90, so its value $5/3$ (between the no-signal value 2 and the full-revelation value 1) is reached by no bounded number of fair bits; this identification is by comparison of the two declared scopes, since the ST-M90 manifest does not list ST-M41. The garbling $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ in the repository's Blackwell separation note cannot be implemented with any bounded number of fair bits; its exact expected price is $8/3$, and one ternary toss pays it (ST-M93).

Which nonclaims are now priced. For finite rational one-shot shared-codebook games satisfying (A1) to (A4), ST-T310–ST-T312 price the non-dyadic seed generation left open by ST-T95 ("does not price exact non-dyadic random-seed generation") completely for a fixed budget, and for a variable budget exactly when $\mathcal{O}$ is a singleton (as an attained minimum in general). Against ST-T109, which notes that support size is not the random-bit cost, support size enters only the rounding constant $\lfloor \sigma/2 \rfloor$; the cost is governed by dyadicity of $\mathcal{O}$ and by κ. For the lanes ST-T171, ST-T174, ST-T182 and ST-T184 the repository records only a remark: *if* the relevant public-mixing game satisfies (A1) to (A4) and its optimal set has no dyadic point, its value is attained by no bounded-bit seed. Those matrices were not rechecked, and the random-bit nonclaim of ST-T152 is not closed.

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